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September 28, 2026

Editor-in-Chief

Journal of Computer-Aided Molecular Design

Springer Nature

Dear Editor,

We submit the manuscript **“Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis”** for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

Scientific Contribution. This work addresses a methodological challenge in computational antimalarial discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing drug candidates from African natural-product-inspired chemical space. Our approach maintains clear boundaries between computational evidence and biological activity claims—a distinction critical for reproducible hypothesis generation in computer-aided molecular design.

The workflow integrates three computational layers:

1. **Chemical-space expansion:** A hybrid library of 65 856 molecules generated from 396 African natural products and 454 synthetic antimalarial compounds, yielding 19 913 computationally prioritized synthesizable candidates (92.6 % structurally novel, 69.3 % scaffold recovery).
2. **Target-anchored docking:** A 17-member polypharmacology-oriented cohort was evaluated against four mechanistically diverse antimalarial targets (PfDHFR, PfCRT, PfClpP, PfATP4) using AutoDock Vina with target-specific structural anchors. All 68 candidate–target pairs passed geometric quality criteria, generating a complete 17×4 scoring matrix.
3. **Per-target resistance-resilience scoring (RRS):** Mutation resilience was quantified separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels. The cohort yielded seven A*, four B, and six C RRS profiles (range 73.2 %–115.6 %). Non-binding wild-type targets were excluded from RRS denominators to prevent artifactual ratios—a design choice that distinguishes this framework from pooled-target approaches.

Validation and Reproducibility. The docking protocol was validated through three complementary approaches: (i) external DEKOIS 2.0 benchmark demonstrating near-chance enrichment for PfDHFR (ROC-AUC 0.45), constraining interpretation of absolute scores while preserving relative within-target rankings; (ii) MMV Malaria Box enrichment showing strong performance for the consensus protocol (ROC-AUC 0.924–1.000); and (iii) successful redocking of reference ligands (RMSD < 2.0 Å for four of five ligands). A retrospective panel of five approved antimalarials against the same mutant sets reproduced the null character of the corrected PfCRT channel, confirming that docking-derived RRS values require experimental validation before informing resistance-circumvention claims.

Complete computational reproducibility is ensured through our Zenodo deposit (<https://doi.org/10.5281/zenodo.22696778>, MIT license), which contains: source code, candidate manifests,

receptor and ligand files, grid configurations, raw docking poses, RRS protocols, and machine-readable tables with SHA256 verification.

Relevance to JCAMD Scope. This work offers several contributions relevant to the journal’s focus on computer-aided molecular design:

1. **Computational efficiency:** The centroid-based library reduction achieves 99.3 % cost reduction (from 2600–5300 CPU-hours to under 20 CPU-hours for the four-target panel), making systematic multi-target and mutation-panel screening accessible to research groups with limited computational infrastructure—particularly relevant for endemic-region institutions where 94 % of malaria cases occur.
2. **Methodological framework:** The separation of target breadth from mutation resilience, combined with target-specific evidence reporting rather than uncalibrated cross-target averaging, provides a reusable template for hypothesis generation in resistance-challenged therapeutic areas.
3. **African natural product chemistry:** This underexplored chemical space is systematically characterized with quantitative novelty metrics and scaffold-retention analysis, contributing to diversity-oriented drug discovery beyond established synthetic compound libraries.
4. **Statistical rigor:** Statistical power analysis, Bonferroni correction, permutation tests, and explicit null controls demonstrate how computational hypotheses can be generated with transparent uncertainty quantification.

Evidence Boundaries. We emphasize that this study does not claim experimental affinity measurements, confirmed simultaneous target engagement, biological resistance circumvention, or measured IC_{50}/EC_{50} values. The manuscript produces: (a) four-target docking profiles from which polypharmacology hypotheses can be generated; (b) per-target RRS values summarizing docking-score ratios on mutation panels; and (c) exploratory cross-metric associations quantified with appropriate statistical correction. These layers are kept separate rather than conflated, making the resulting hypotheses experimentally falsifiable.

Author Contributions and Declarations. This manuscript is original work, is not under consideration elsewhere, and has been approved by all co-authors. We declare no competing financial interests. No external funding was received for this computational study. AI-assisted tools were used during code development and data-analysis scripting; all output was reviewed and edited by the authors, who take full responsibility for the content.

We believe this work exemplifies the rigorous computational methodology and clear evidence boundaries valued by the *Journal of Computer-Aided Molecular Design* community. We would be pleased to provide any additional information or clarification you may require.

Sincerely,

Myke Vital Sao Temgoua
(on behalf of all co-authors)